École Polytechnique – HEC Paris | Master Data & Finance

Risk and Performance of a Dynamic Markowitz Portfolio on CAC 40 Constituents

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Abstract

We construct and evaluate a daily-rebalanced Markowitz portfolio on ten diversified CAC 40 constituents using one-minute OHLC data (September 2022 – February 2023, $n = 120$ trading days). Expected returns are proxied by AR(1) forecasts built from lag-1 autocorrelations $\hat{\rho}_j \in [-0.168, 0.123]$. The unconstrained optimisation produces extreme leverage ($\sum |\Omega_j| \approx 22$) that destroys capital: the portfolio ends at -1271.82 (base 100) versus $+122.89$ for the equal-weight benchmark. Ex-ante Sharpe is 0.96 (annualised) vs. ex-post 0.44; UPR = 0.32; maximum drawdown = -467%. The exercise illustrates the dangers of combining unconstrained Markowitz with weak forecasts and full-sample covariance estimation.

1 Dataset and Investment Universe

The raw data consist of minute-frequency OHLC prices for all 40 CAC 40 constituents stored as `.xlsx` files. The analysis window 13 September 2022 – 28 February 2023 yields $n = 120$ trading days and $n - 1 = 119$ daily returns per stock. Daily closing prices are obtained by retaining the last intraday quote of each session.

We select ten liquid, sector-diversified stocks (Table 1). Liquidity is confirmed by row counts $\geq 58\,000$ minute-bars (the bottom quartile of the full CAC 40 falls below $\sim 40\,000$). At most one stock is taken per sector to avoid systematic factor concentration.

Table 1: Investment universe: ten CAC 40 constituents.

Ticker	Company	Sector	Rows
MC	LVMH	Luxury Goods	61 371
BNP	BNP Paribas	Banking	60 911
AIR	Airbus	Aerospace & Defence	60 769
CS	AXA	Insurance	60 150
ORA	Orange	Telecoms	59 821
TTE	TotalEnergies	Energy	59 410
SU	Schneider Electric	Industrials	59 394
DSY	Dassault Systèmes	Enterprise Software	59 276
SAN	Sanofi	Healthcare	58 861
MT	ArcelorMittal	Materials	58 454

2 Portfolio Construction

2.1 Daily Returns

Arithmetic daily returns $R_{i,j} = (P_{i,j} - P_{i-1,j})/P_{i-1,j}$ form a 119×10 matrix. Returns are approximately centred at zero with occasional large outliers (Figure 1); concurrent spikes reflect systematic market events.

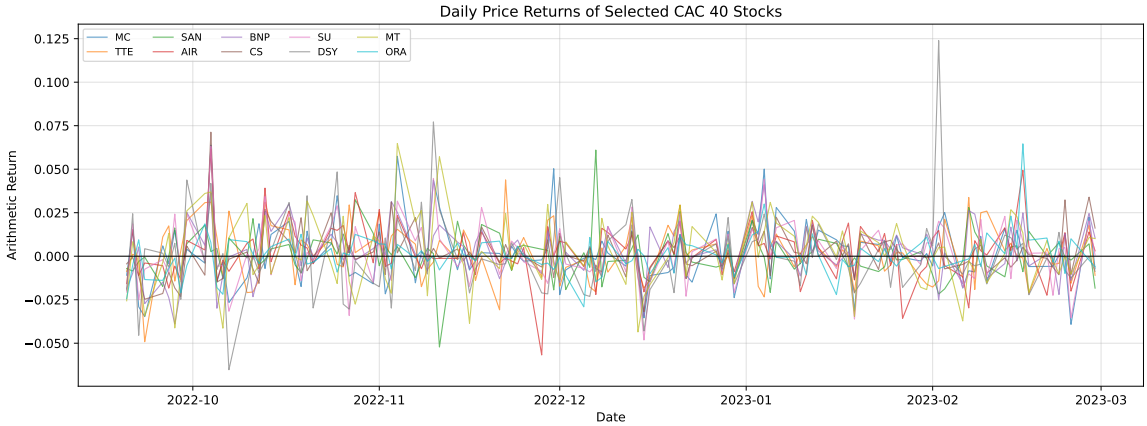

Figure 1: Daily arithmetic price returns for the ten selected stocks.

2.2 Correlation and Covariance

Figure 2 displays the sample correlation and covariance matrices. All 45 pairwise correlations are positive (average 0.29), reflecting a shared European equity systematic factor. The strongest links are MC–SU (0.74) and BNP–CS (0.64); Sanofi and Orange show the weakest co-movements, confirming their role as defensive diversifiers.

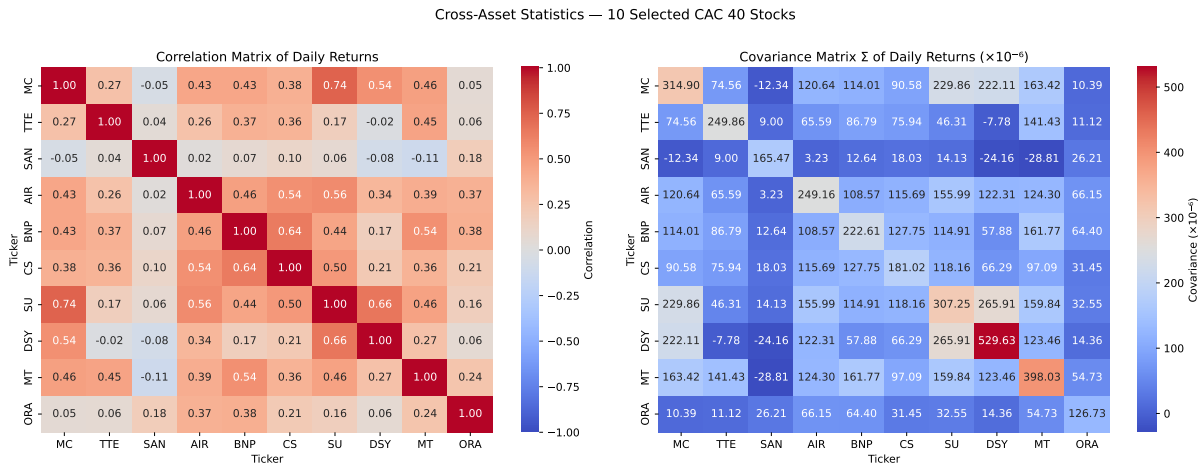

Figure 2: *Left*: sample correlation matrix. *Right*: covariance matrix ($\times 10^{-6}$). All pairwise correlations are positive.

2.3 Lag-1 Autocorrelation

The lag-1 autocorrelation $\hat{\rho}_j$ is estimated for each stock (Table 2, Figure 3). Values lie in $[-0.168, 0.123]$: seven stocks show mean reversion (negative, consistent with bid–ask bounce) and three show slight momentum. Despite their small magnitude, these are the sole AR(1) predictors used for expected-return forecasting.

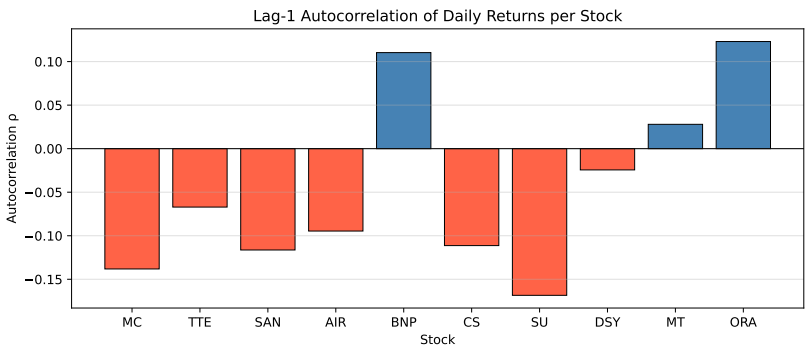

Figure 3: Lag-1 autocorrelations $\hat{\rho}_j$. Negative: mean reversion; positive: momentum.

Table 2: Lag-1 autocorrelation $\hat{\rho}_j$ per stock.

Ticker	$\hat{\rho}_j$	Ticker	$\hat{\rho}_j$
MC	-0.138	BNP	$+0.110$
TTE	-0.067	CS	-0.111
SAN	-0.116	SU	-0.168
AIR	-0.095	DSY	-0.024
MT	$+0.028$	ORA	$+0.123$

2.4 AR(1) Projections and Markowitz Weights

Under a zero-mean AR(1), the expected return at day $i+1$ is $\hat{R}_{i+1,j} = \hat{\rho}_j \cdot R_{i,j}$. The Markowitz weight vector (unit-sum constraint, no leverage cap) is:

$$\Omega_{i+1} = \frac{\Sigma^{-1}\hat{R}_{i+1}}{\mathbf{1}'\Sigma^{-1}\hat{R}_{i+1}}. \tag{1}$$

Because $|\hat{\rho}_j| \ll 1$ the denominator $\mathbf{1}'\Sigma^{-1}\hat{R}_{i+1}$ is near zero on most days, producing extreme leverage (Figure 4): time-averaged $\sum |\bar{\Omega}_j| \approx 22$, with individual mean weights ranging from -3.86 (AXA) to $+3.54$ (LVMH).

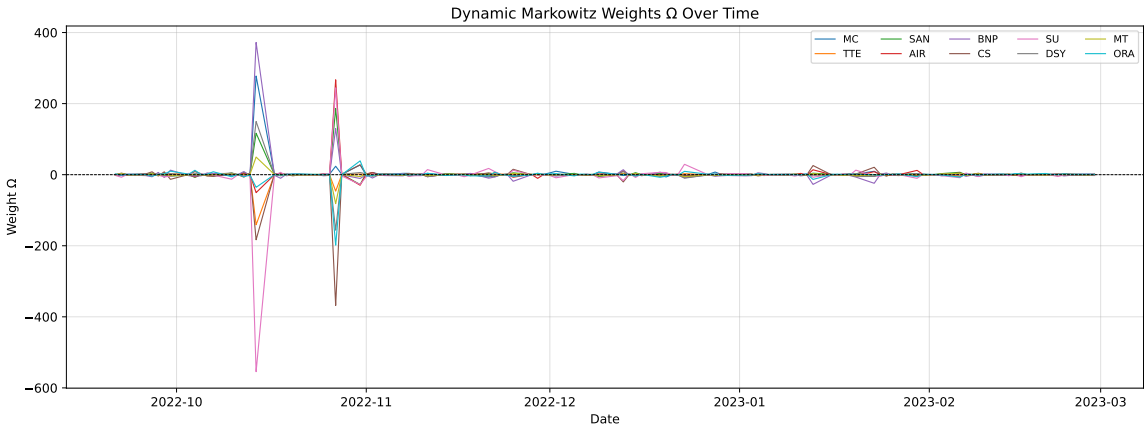

Figure 4: Daily Markowitz weights over the investment period. Extreme excursions arise when the AR(1) projected return is near zero.

2.5 Realised Portfolio Returns and Look-Ahead Bias

The daily realised portfolio return $r_{p,i+1} = \Omega'_{i+1}R_{i+1}$ is compounded into a cumulative value $V_{i+1} = V_i(1 + r_{p,i+1})$ (Figure 5). The portfolio ends at $-\mathbf{1271.82}$ (base 100) versus $+122.89$ for an equal-weight benchmark: extreme leverage turns even occasional forecast errors into losses exceeding available capital.

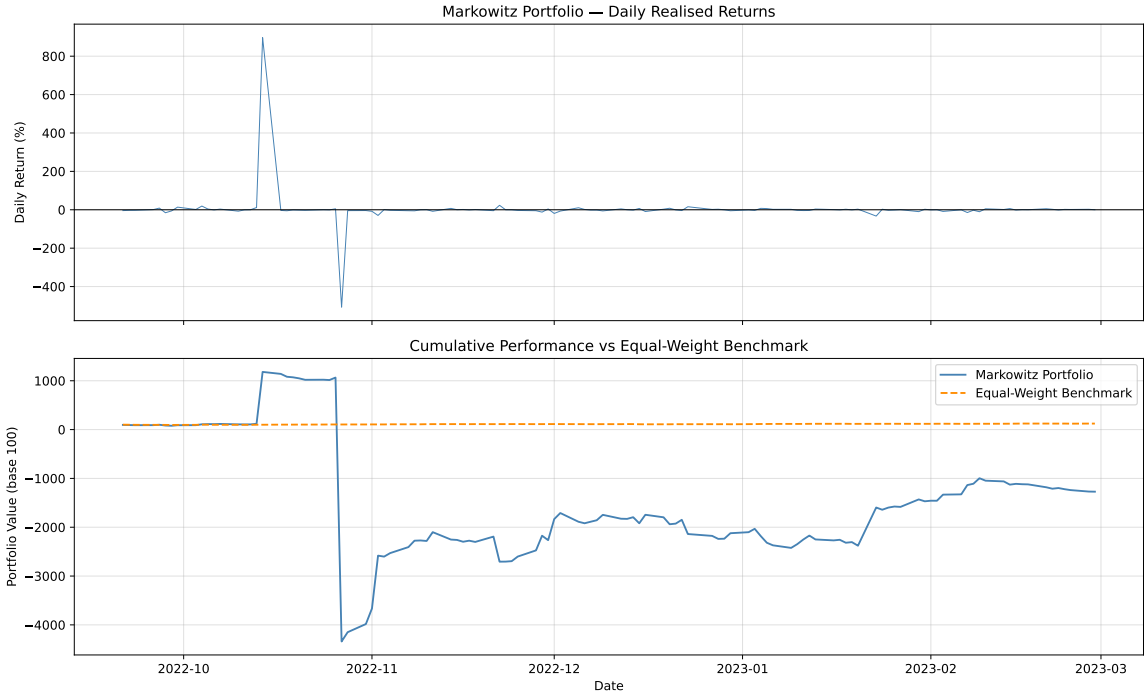

Figure 5: *Top*: daily portfolio returns (range [−507%, +897%]). *Bottom*: cumulative value; equal-weight benchmark shown in orange.

Look-ahead bias. Σ in (1) is estimated on the *full* return sample, including data future to any given trading day — a textbook look-ahead bias. Here the bias cannot offset the absence of a real forecasting edge; a rolling-window Σ would reduce bias without solving the fundamental leverage problem.

3 Risk and Performance Evaluation

3.1 Sharpe Ratio

Sharpe $= \bar{r}_p/\sigma_p$ (risk-free rate $= 0$). We report ex-ante (using projected returns $\hat{r}_{p,i+1} = \Omega'_{i+1}\hat{R}_{i+1}$) and ex-post (using realised returns).

Table 3: Sharpe ratio of the Markowitz portfolio.

Variant	Daily SR	Annualised ($\times\sqrt{252}$)
Ex-ante	0.0604	0.9595
Ex-post	0.0277	0.4394

The ex-ante SR of 0.96 is by construction the maximum achievable from the AR(1) signal. The realised SR drops to 0.44: only $\approx 46\%$ of the constructed signal survives in actual returns, a direct consequence of the low autocorrelation magnitudes. Both figures correspond to a daily standard deviation $\approx 97\%$ (extreme leverage), making conventional Sharpe interpretation problematic.

3.2 Upside Potential Ratio

$$\text{UPR} = \frac{\mathbb{E}[\max(r_p, 0)]}{\sqrt{\mathbb{E}[\min(r_p, 0)^2]}} = \frac{0.212}{0.655} = 0.323.$$

The UPR well below 1 confirms heavily left-skewed returns: average losses (0.655 per losing day) outweigh average gains (0.212) by a factor of 3. Leverage amplifies forecast errors disproportionately — when the AR(1) signal is wrong on a large market move the loss is multiplied by the full 22× leverage.

3.3 Maximum Drawdown

$$\text{MDD} = -467.4\%, \quad \text{Peak: 14 Oct 2022}, \quad \text{Trough: 27 Oct 2022}.$$

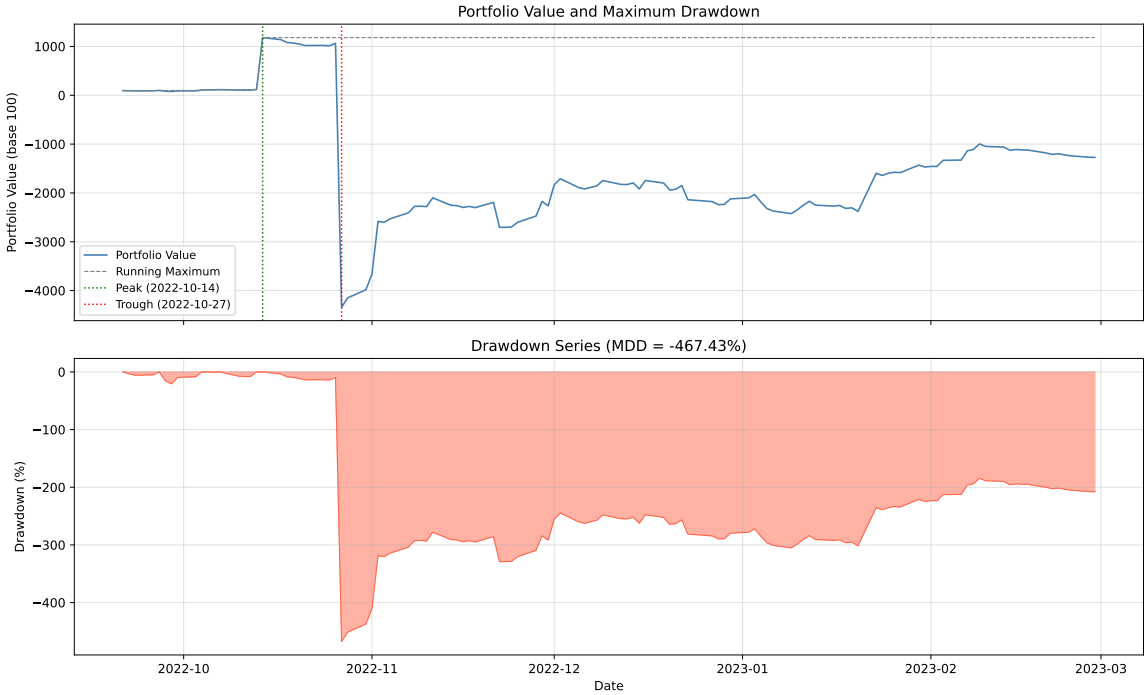

Figure 6: Cumulative value (top) and drawdown series (bottom). Peak (+1181, green) and trough (−4340, red) are marked.

The MDD exceeds −100% because the portfolio value becomes deeply negative. The critical two-week window (14–27 Oct 2022) coincided with peak Q4 2022 European market stress (BoE emergency gilt intervention, ECB tightening climax, energy-price spike). Large intraday reversals during this period caused the AR(1) projections to be systematically wrong, and extreme leverage converted these errors into catastrophic losses.

3.4 Additive Volatility Contributions

$$\sigma_p = \sum_{j=1}^{10} C_j, \qquad C_j = \frac{\bar{\Omega}_j \cdot (\Sigma\bar{\Omega})_j}{\sigma_p}, \tag{2}$$

evaluated at the time-averaged weight vector $\bar{\Omega}$ (Table 4, Figure 7).

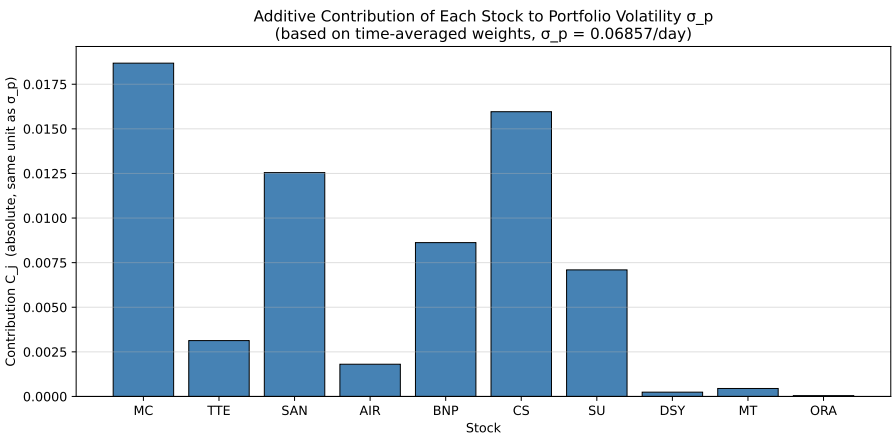

Figure 7: Additive contribution C_j per stock to portfolio daily volatility $\sigma_p = 0.0686$ (all positive, even short positions).

Table 4: Volatility decomposition at time-averaged weights ($\sigma_p = 0.0686$/day, annualised $1.089 = 108.9\%$/year).

Ticker		$\bar{\Omega}_j$	C_j	Share
MC	LVMH	+3.538	0.01868	27.24%
CS	AXA	−3.855	0.01596	23.27%
SAN	Sanofi	+2.773	0.01255	18.30%
BNP	BNP Paribas	+3.055	0.00862	12.57%
SU	Schneider Elec.	−3.261	0.00709	10.35%
TTE	TotalEnergies	−1.338	0.00313	4.57%
AIR	Airbus	+1.973	0.00181	2.63%
MT	ArcelorMittal	−0.362	0.00045	0.65%
DSY	Dassault Syst.	−0.143	0.00024	0.35%
ORA	Orange	−1.378	0.00004	0.06%
	Total		0.06857	100.0%

Annualised portfolio volatility is 108.9% vs. ≈ 12–16% for individual stocks. Risk is concentrated in three names (MC + CS + SAN = 68.8%) and, crucially, **all** $C_j > 0$ including short positions. Because all pairwise correlations are positive, short positions still move together with the portfolio in the direction of their own weight — there is no true hedging effect.

4 Conclusion

Seven key findings:

(i) All pairwise correlations positive (avg 0.29); Sanofi and Orange are the weakest-linked, most defensive names.

(ii) AR(1) autocorrelations in $[-0.168, 0.123]$: seven mean-reverting, three momentum. Signals are near-zero and noise-dominated.

(iii) Unconstrained Markowitz amplifies near-zero forecasts into 22× leverage, the root cause of all subsequent failures.

(iv) Ex-ante SR 0.96 vs. ex-post SR 0.44 (annualised): only 46% of the optimised signal materialises in realised returns.

(v) UPR = 0.32: losses outweigh gains 3-to-1 in a left-skewed, leveraged distribution.

(vi) MDD = −467%, portfolio ends at −1272 (base 100); worst episode triggered by Q4 2022 European market stress.

(vii) All ten volatility contributions are positive: no genuine diversification in an all-positive-correlation universe under unconstrained weights.

Remedies. Two straightforward improvements would transform this from a mathematical exercise into a deployable strategy: (i) **leverage/position constraints** ($|\Omega_j| \leq 0.2$) to prevent weak signals from generating extreme bets; and (ii) **Ledoit–Wolf covariance shrinkage** to dampen $\|\Sigma^{-1}\hat{R}\|$ when $\hat{R}$ is small.

Code: all computations are in the accompanying notebook `main.ipynb`.